

If you think I am making too big a deal about a cup of tea, please bear with me. Imagine that you have ₹15 with you today. You get a cuppa for ₹7, a mint for ₹1 and are left with ₹7 with which you are going to conduct an experiment. You invest the remaining ₹7 and forget about it. Twenty years later, you suddenly remember and net the profit. Excited beyond measure, you head to the same chai shop conjuring past memories.

In-order to buy the same cuppa, you would need a post-tax return of *at least* 10% matching the estimated chai price of ₹47. And it is only an estimate! Often, life doesn't pan out like a spreadsheet. Perhaps, due to overpopulation and lack of resources, a cuppa could cost much more, or maybe someone invented artificial tea that you could grow from your homes!

When you extrapolate this experiment with indispensable monthly expenses – groceries, vegetables, petrol, cooking gas, electricity, phone, internet, cable maybe costing you 15k currently, the cost of each item is likely to increase in the future, but not at the same linear rate. A hard look at my family's ledger report reveals that the price of rice has only increased by about 3-4% over a 20+ year period (1995-2017). However, the price of vegetables has gone up by 9-10%, while electricity is up by about 9%. The overall rate at which our monthly expenses have grown is roughly 8-9% yearly.

This means that if I spend ₹15,000 in January 2017 towards monthly expenses, I should safely expect to spend 16,200 in Jan 2018

~ 17,500 in Jan 2019

~ 18,900 in Jan 2020

~ 69,900 in Jan 2037.

Of course, the expenses may not *actually* increase 8% each year. It may be the same (15,000) for the next 3 years and in 2020 shoot by 36% (~ 20,400). If you look back, the year on year increase is still 8%. This R (8%) is known as the *compounded annualized growth rate* or CAGR. The real-life representation follows this kind of a model, wherein prices don't increase for a considerable period and then, shoot up vertically. When you spread it across and calculate CAGR, you'll note that expenses notably double every 9 years.